

Walter Teitelbaum

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Education

UC Santa Cruz - Santa Cruz, CA

BS, Mechatronics, Robotics, and Automation Engineering

Objective

Hands-on automation and mechatronics engineer who owns custom robotic manufacturing/test equipment from requirements and controls architecture through design, fabrication, qualification, and production deployment. Driven to solve hard technical and organizational problems that deliver real business impact. Strongest in first-principles design, industrial motion control, safety-interlocked equipment, Python-based control software, practical electrical/mechanical integration, and manufacturing test systems. I have built automation systems in a startup setting and bring additional perspective from Boeing on production risk, manufacturing scale, and process-driven execution.

Skills and Expertise

Industrial Communication & I/O: EtherCAT, CANopen, CiA-402 / DS402 motion devices, PDO/SDO mapping, SOEM, Modbus TCP, CAN, serial communication, TCP/UDP, PLC discrete I/O, PLC analog output scaling, relay/contacter control, interlock logic, watchdogs, heartbeat monitoring, communication-loss handling

Robotics & Motion Control: Servo and stepper drive integration, 5-axis robotic systems, gantry/linear/rotary/gimbal axes, homing routines, overtravel recovery, velocity/acceleration/deceleration tuning

Safety, Risk & Commissioning: FMEA, hazard analysis, fail-safe / fail-closed logic, E-stop integration, machine barriers, safe-stop behavior, drive fault recovery, fault-state machines, commissioning checklists, qualification testing

Software & Test Automation: Python, asyncio, FastAPI, REST APIs, WebSocket telemetry, Pydantic, Typer, YAML configuration, structured logging, hardware abstraction layers, state machines, pytest, hardware-interface mocking, CI/CD, Git, Bitbucket Pipelines, mypy, Ruff, pre-commit, thoughtful prompt engineering with Codex and Claude Code

Test, Metrology & Data Acquisition: Optomechanical test and measurement system design, scientific data engineering, NumPy, xarray, NetCDF4, measurement validation, metadata provenance, test artifact generation

Mechanical, Electrical & Fabrication: SolidWorks, Onshape, Bambu Studio, Formlabs PreForm, Makera CAM, CNC Milling, Resin and FDM 3D Printing, fixture design, soldering, sheet metal fabrication, material handling and rigging, carpentry, industrial control wiring, wiring harness fabrication, high-voltage EV battery testing

Manufacturing Systems & Leadership: NPI equipment ownership, manufacturing at scale, industrial engineering, process engineering, production risk mitigation, engineering audit planning, Jira, Smartsheet, Notion, supplier/vendor coordination, procurement, technical documentation, cross-functional technical leadership

Professional Experience

Lumotive - Member of Technical Staff - Test Automation Engineering June 2025 - Present

- Delivered a custom 5-axis robotic LiDAR characterization platform from requirements definition through commissioning, qualification, and operator deployment.
- Designed and integrated complex automation systems combining servo/stepper drives, EtherCAT/CANopen networks, PLC I/O, and sensor suites.
- Developed reusable Python control software featuring safety state machines, REST/WebSocket telemetry, and automated CI/CD testing via hardware mocks to minimize deployment risk.
- Programmed advanced motion profiles (S-curve tuning, collision avoidance) and fail-safe logic (E-stops, interlocks, and redundant safety-gated APIs).
- Architected automated production test lines for DToF/IToF LiDAR PCBAs, incorporating robotic active optical alignment, spectral characterization, and calibration workflows.
- Led multidisciplinary engineering teams (Optics, EE, FW, ME) to resolve complex integration issues, manage procurement, and author production-ready documentation.

Boeing - Renton, WA; Manufacturing Engineer II January 2024 - May 2025

- Worked closely with members of the technical fellowship to architect and plan internal engineering audits for a new initiative focused on eliminating risk in the production system.
- Developed a complex work management environment in Smartsheet to assist teams in identifying follow-on actions and tracking them to completion.

Zero Motorcycles - Scotts Valley, CA; Electrical Engineer II August 2021 - December 2023

- Maintained and continuously improved the production line for a high-power battery assembly.
- Wrote several custom Python tools to interact closely with CAN devices and to facilitate battery charging and telemetry, as well as motor encoder testing and quality.
- Responsible for assembling production-quality automotive wiring harnesses for use on the line.

Formula Slug FSAE Chapter - Santa Cruz, CA; Director of Finance

Aug 2019 - Dec 2022

- Built an electric racecar from the ground up, from CAD modeling to battery design, HV electrical system integration to aerodynamics, and data acquisition.
- Designed several components for the racecar's steering, braking, and tractive system monitoring subsystems. Worked closely with suppliers and local organizations to procure consumer off-the-shelf parts and stock for in-house fabrication.

UC Berkeley AUTOLAB - Berkeley, CA; Research Intern

July 2021 - Aug 2022

- Designed a mechatronic robot arm for AlphaGarden: an autonomous garden that uses an ML-based simulation platform to plant, prune, and irrigate living plants in a physical testbed.
- Collaborated on and submitted multiple entries to [ICRA](#), the IEEE International Conference on Robotics and Automation, and [CASE](#), the International Conference on Automation Science and Engineering. Finalist for best conference and application paper at CASE 2022.

FIRST Tech Challenge Team - Piedmont, CA; Founder & Team President

Jan 2015 - Sep 2018